Real Time Smart Energy Management System

Optimizing power source selection
with real time sensing and Time of Use pricing.

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Design Concepts and Applications

Our project implements an intelligent energy management system intended to monitor electrical conditions in real time, enabling autonomous selection between grid power and a battery-backed renewable source.

This system utilizes measured voltage and current data, overall battery condition, and provided time of use pricing to determine which source is the most appropriate to supply the current load demand.

The objective: Reduce energy expense to the consumer through optimization during peak-rate periods while ensuring reliable operation and safe power source switching. Our finalized implementation evaluates the current load, grid, and renewable status, using a single microcontroller to select the appropriate source to pass through towards the connected load.

Why its meaningful

- Energy is getting more expensive, reflected during peak demand periods
- Dynamic source selection:
  - Increase efficiency
  - Reduce operation costs
- Applications in residential households, power backup systems, and renewable energy storage applications

Electrical Engineering related concepts

- Embedded control systems
- I2C Serial Communication
- Power monitoring
- Low voltage circuit design
- Source-aware decision logic
- Real-time telemetry
- Remote cloud network integration

Technical Design Objectives

Measurable and performance-aware EMS performance targets

Monitoring

- Active voltage/current sensing on load, grid, battery
- Update telemetry dashboard every second.
- Fan speed update every 250 ms.
- Values must be within 1 second of real time.

Control Logic

- Automatic source selection using embedded time-of-use pricing.
- Battery discharge only if;
  - SOC ≥ 35%, Voltage ≥ 12.20V to enter battery mode
  - SOC ≥ 25%, Voltage ≥ 12V to remain on battery

Protection / System Operation

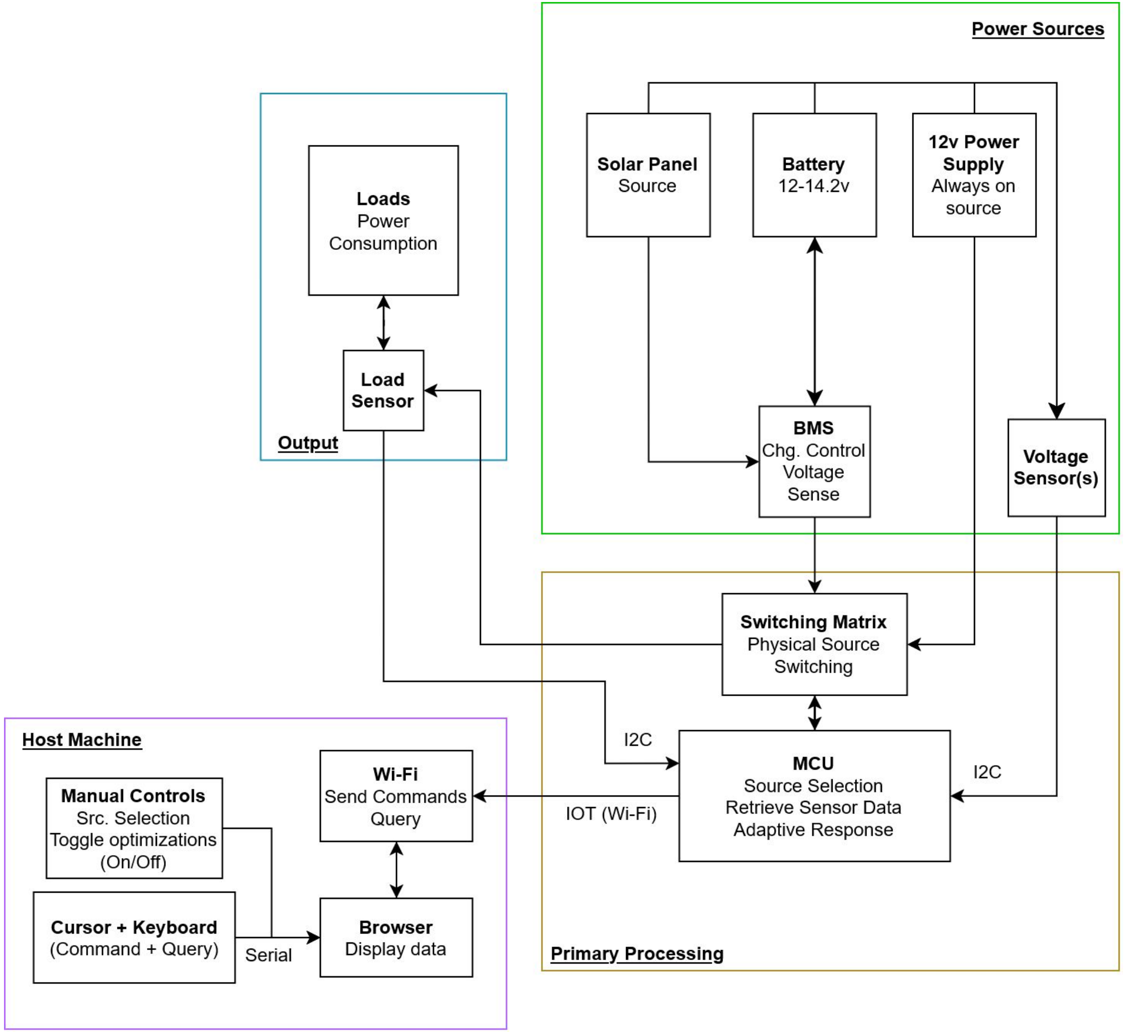
- Safe relay transfer with:
  - 150 ms break-before-make delay
  - 600 ms min. automatic dwell
  - 2500 ms min. manual dwell
- Detect unsafe conditions:
  - relay conflict
  - overcurrent above our threshold (~6A)
- Cloud status and telemetry display through Arduino IoT Cloud

The following objectives ensure measurable, safe, and stable EMS performance suitable for a real time demonstration.

System Description

Block Diagram

- **Power Sources:**
  - Renewable, Always on “Grid”
  - Managed renewable energy
- **Primary Processing:**
  - Microcontroller
  - Reactive source selection
  - Active data monitoring + optimization
- **Host Machine:**
  - Interactive dashboard
  - Status Display + Manual Control
- **Output:**
  - “Household” devices
  - Real time load measurements



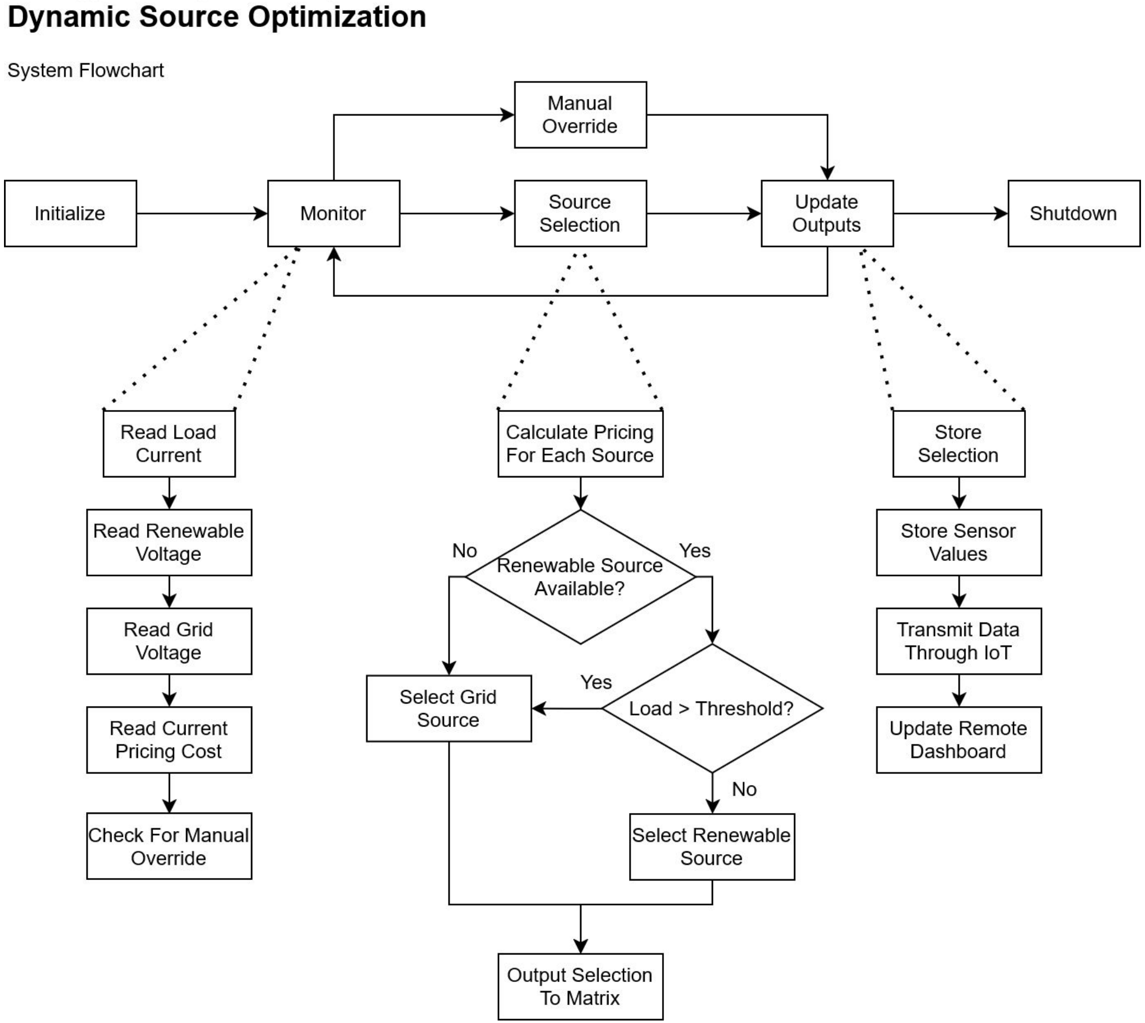
System Description

Logical Flowchart

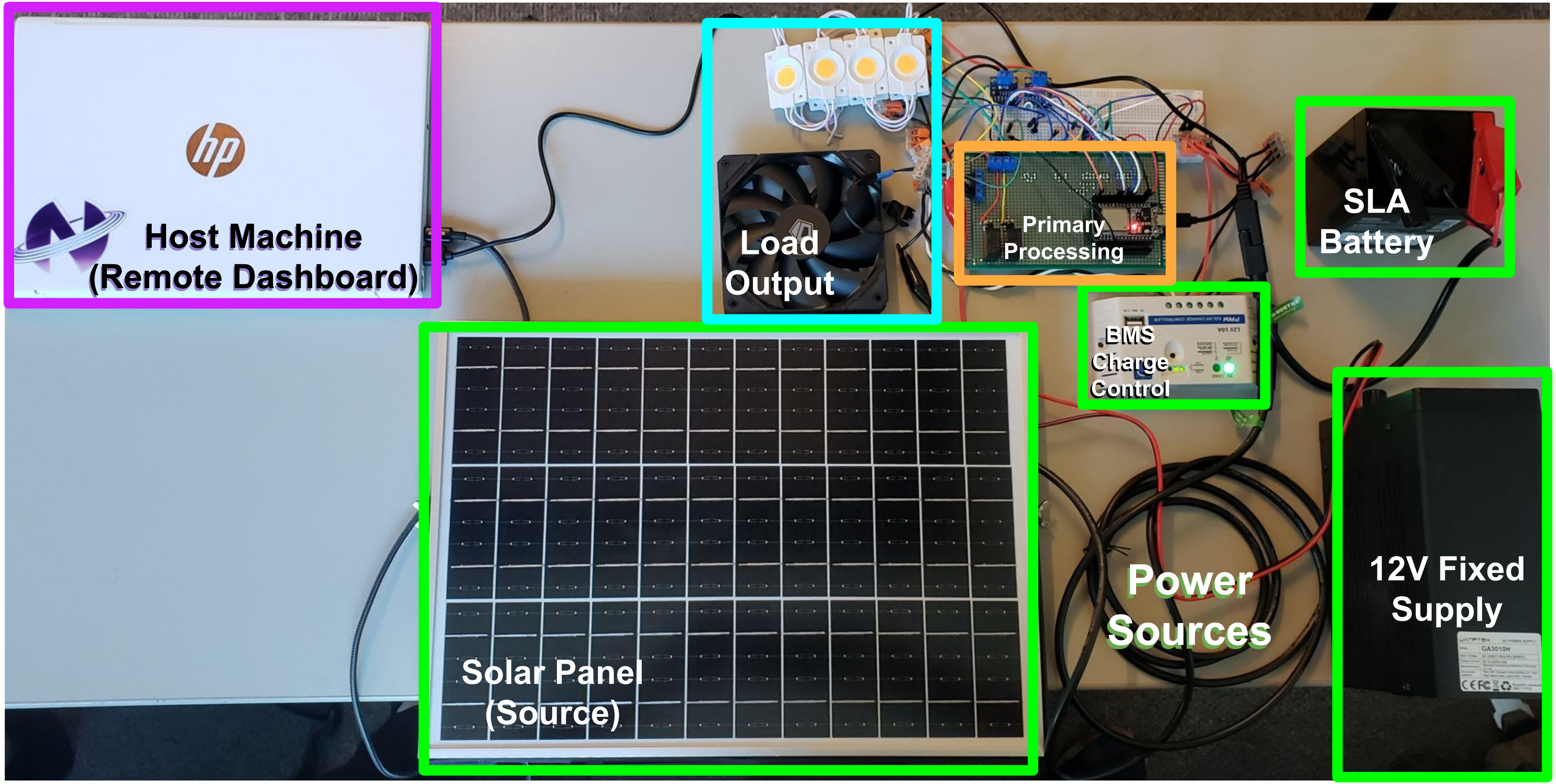
Basic Principle:

1. Gather and read information
2. Select and switch source
3. Update dashboard display

- Manual override at any time
- Time of Use rates are stored locally
- Decisive and consistent selection (Oscillation prevention)



General Component Layout



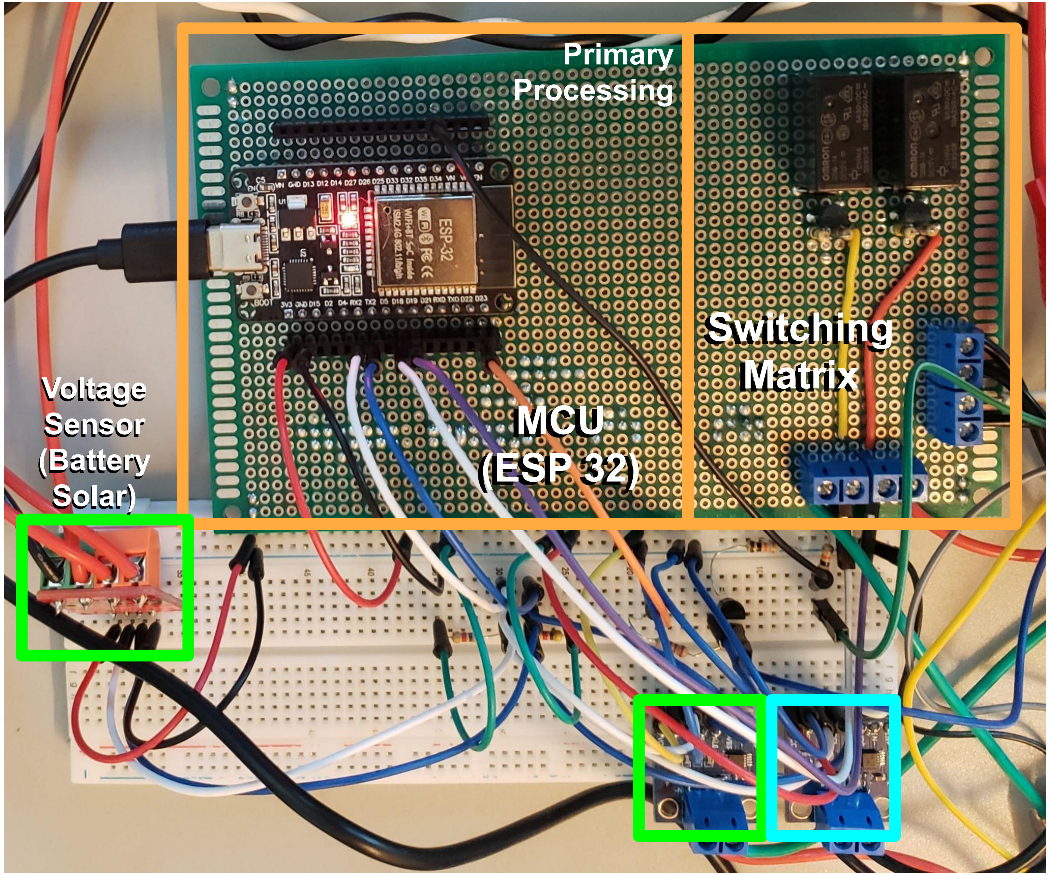
General Component Layout

- Implementation of “Primary Processing” module:
  - Single Microcontroller (ESP32)
  - 2 source modular switching matrix
  - I2C communication, external sensors

- External Sensor Modules:
  - INA226 Voltage/Current sensors
  - x2 Input for current source status
  - x1 Feedback for current load demand

Angelo: Designed the switching matrix, wiring and assembly of the whole system, and responsible for the EMS algorithm, UI dashboard and cloud/IoT integration.

Christopher: Worked on the hardware parts of the system, soldering the relay matrix circuit and the esp32 onto the perfboard, and soldering/modifying the INA226s to meet our specs

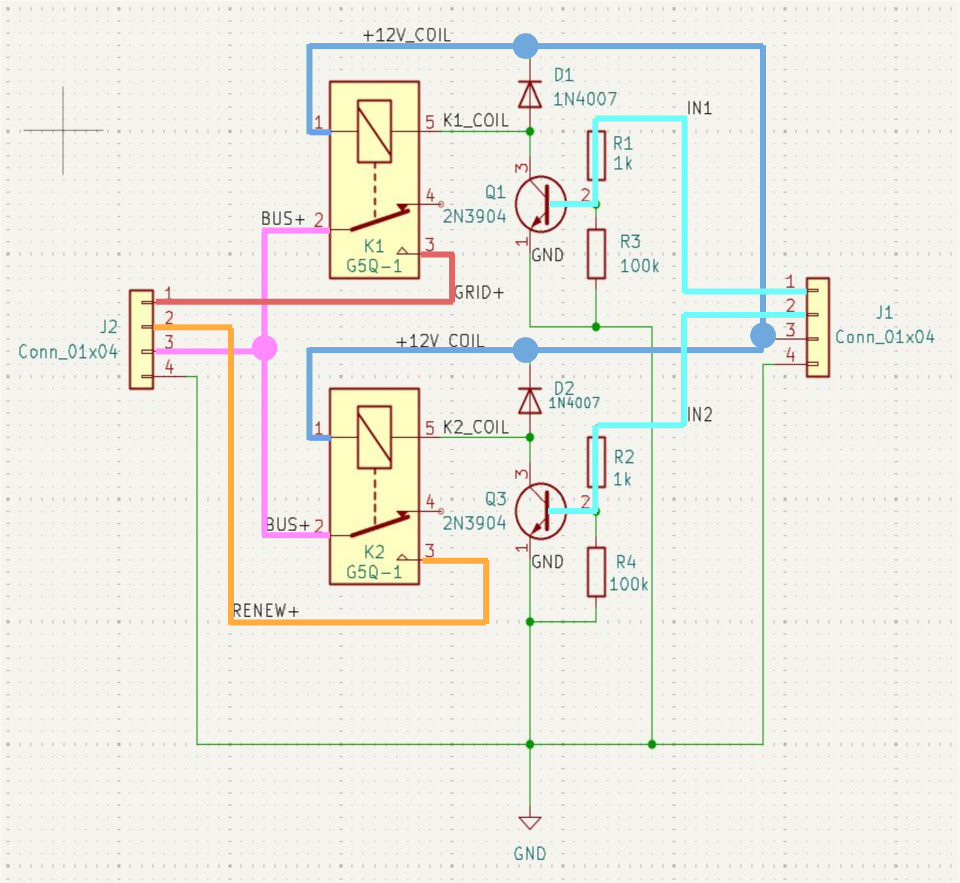


Switching Matrix

- Implementation as a discrete circuit
  - Based around series of 12V relays
  - Actuated through transistor switching
  - Direct connection to Microcontroller

- **Inputs**
  - Supplemental relay power (+12V)
  - x2 Selection signal (0-5V)
  - Direct “Grid” input (+12V)
  - Direct “Battery” input (+12-14V)

- **Outputs**
  - Load source (12V) from either:
    - “Grid” output (+12V)
    - “Battery” output (+12-14V)
  - Shared ground (0V) power rail



EMS Pseudo Code

START EMS AUTO DECISION LOOP

Every 1 second:

1. Update simulated clock
- Advance sim time based on sim speed
- Update current hour (simHour)

2. Determine current price band
IF tariffProfile == Riverside summer:
peak hours = 16 to 20
peak price = 0.55
off-peak price = 0.22
ELSE IF tariffProfile == Chicago winter:
peak hours = 7 to 9 and 17 to 19
peak price = 0.22
off-peak price = 0.16

currentPrice = price at simHour
inPeak = whether simHour is inside peak period

3. Read battery condition
batteryVoltage = measured battery voltage
batterySOC = estimated battery state of charge

4. Decide whether battery path is allowed
IF currently using GRID:
battery is allowed only if
batterySOC >= 35%
AND batteryVoltage >= 12.20 V
ELSE IF currently using BATTERY path:
battery is allowed to continue if
batterySOC >= 25%
AND batteryVoltage >= 12.00 V

battOk = result of this hysteresis check

5. Time-of-use source selection rule
IF battOk == true AND inPeak == true:
candidateRenew = true
ELSE:
candidateRenew = false

Meaning:
- During expensive peak hours, **use battery if battery is healthy**
- During off-peak hours, use grid
- If battery becomes weak, **fall back to grid**

6. Add decision confirmation delay
IF candidateRenew changed since last check:
remember new candidateRenew
store the sim-time when the change happened

IF candidateRenew has stayed unchanged for at least 5 simulated minutes:
latchedWantRenew = candidateRenew

Meaning:
- Do **not switch immediately** on every small change
- Only switch after condition stays stable for 5 simulated minutes

7. Convert decision into requested relay states
IF latchedWantRenew == true:
relay1_on = false // GRID OFF
relay2_on = true // BATTERY / RENEWABLE PATH ON
ELSE:
relay1_on = true // GRID ON
relay2_on = false // BATTERY / RENEWABLE PATH OFF

8. Apply relay switching safely
- Never allow both relays on at the same time
- Enforce minimum dwell time between switches:
manual mode: 2500 ms
auto mode: 600 ms
- If a change is needed:
a. turn both relays off
b. wait 150 ms break-before-make time
c. turn on only the selected relay

9. Track actual applied relay state
- Save whether GRID relay is physically on
- Save whether BATTERY relay is physically on

10. Final result
IF relay1 is on:
source = GRID
ELSE IF relay2 is on:
source = BATTERY / RENEWABLE PATH
ELSE:

BOTTOM LINE:
```
If (cheap period) {
    use grid
}
else if (expensive period) {
    use battery
}
else {  (unsafe/unnecessary battery use )
    return to grid.
}
```

Technical challenges

Key hardware and control issues encountered during implementation

Sensor / I2C Issues

Challenges:

- Damaged sensors during implementation
- I2C communication inconsistencies
- Overloading sensors causes incoherent data

What we did:

- Damaged components were replaced
- Ease of access / simpler implementation of the sensors to the system.
- Additional pull-up/down for bus signals
- Modified INA226 sensors to measure at higher current conditions

Control / Protection Issues

Challenges:

- Protection from unsafe relay switching
- Overly aggressive fault logic
- Manual control may be inadvertently disabled

What we did:

- Added break-before-make relay switching
- Implemented relay interlock logic so both sources could not be active together
- Revised the fault system to separate warnings from hard faults
- Tuned dwell times and switching behavior for stability

Final Integration / Performance Stability

Challenges:

- Hardware failures slowed validation
- Sensor calibration and shunt variance required debugging
- Reliable design for final demonstration

What we did:

- Sensor Calibration verified through known measurements
- Simplification of hardware where feasible
- Prioritized functional + stable operation over unnecessary features

Design considerations

Practical constraints and engineering interfaces that shaped the final system

Project Constraints

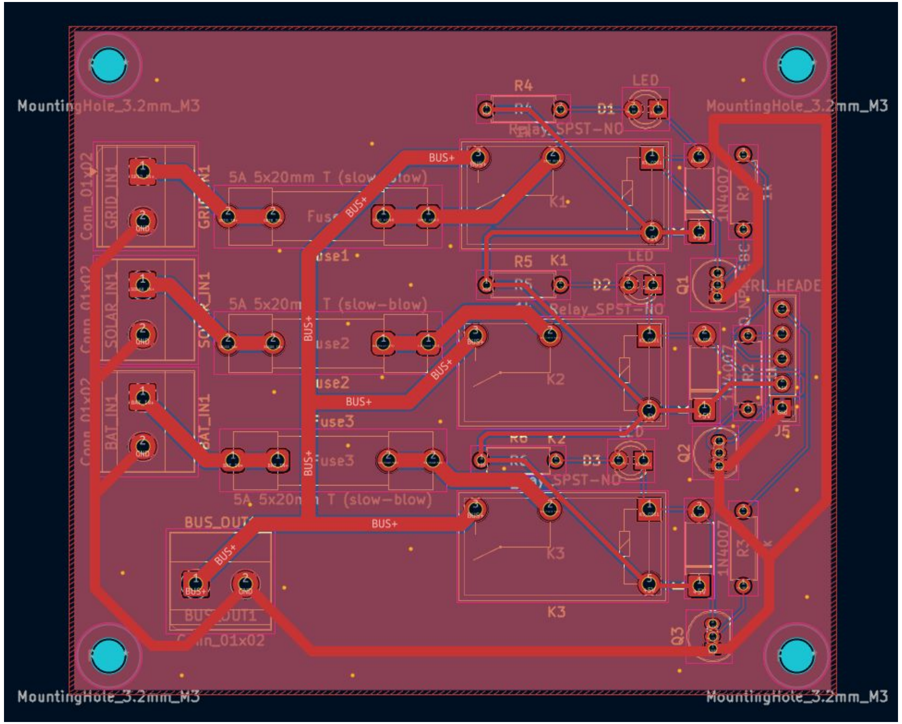
- **Time and integration limits:** Hardware failures and troubleshooting reduced time for expanding features, the final design prioritizes stable sensing, safe switching, and a reliable demonstration.
- **Controller and sampling limits:** The ESP32 handled sensing, relay control, fan control, cloud updates, and decision logic together. The final algorithm was kept lightweight with 1 second telemetry update rate and simple threshold-based control.
- **Cost and implementation tradeoffs:** We favored low-cost, readily available modules and simplified the final sensing configuration to improve repeatability instead of adding more complexity right before demonstration.

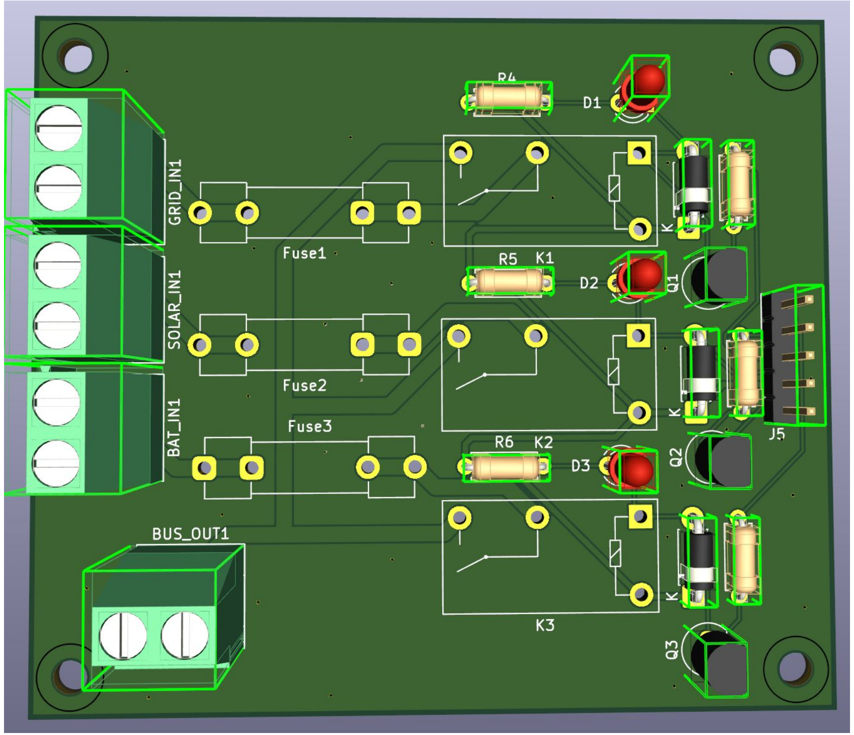
Interfaces / Standards

- **I^2C UM10204**: Used for INA226 communication. This simplified wiring and allowed multiple sensors on one bus, but it also introduced address management and bus-debugging constraints.
- **GPIO / PWM control**: GPIO pins were used for relays and PWM was used for fan control. These interfaces constrained pin assignment and required timing-aware software so sensing and control could run together reliably.
- **IEEE 802.11 Wi-Fi**: Used through the ESP32 for Arduino IoT Cloud connectivity. This enabled remote monitoring, but also meant the dashboard depended on network availability and connection stability

The final EMS design was shaped by practical limits in time, hardware, and controller resources.
In turn, we've emphasized what had been already established at the time of the preliminary testing phase, with basic switching, algorithmic source selection, and basic inter-component communication using I2C.

Preliminary PCB Design





Test report

Our system was tested through sensor verification, real-time telemetry monitoring, manual and automatic relay switching, and time-of-use decision scenarios using the cloud interface and live hardware measurements.

Test	Quantitative Result	Comparison to Design Specification
General sensor communication	Verification of INA226 presence via I2C scan. Telemetry received near 1 second intervals.	Desired Rate: <1000ms update window Measured rate: 800 - 1000ms window
Safe relay switching	Measured source transfer time: <150ms System managed dwell time: 600ms Manual dwell time: 2500ms	Decisive switch, no oscillation. Desired transfer time: <250ms Measured time: <150ms.
Automatic source selection	Renewable used when SOC ≥ 35% & Voltage ≥ 12.3V Remains available until thresholds fall below above.	Renewable active when SOC ≥ 35% & Voltage ≥ 12V.
Failsafe and warning mode	Overcurrent trigger: 5.8A with 1400ms persistence. Overcurrent conditions were met with protection and no damaged equipment.	Overcurrent protection: ≥ 6.0A for ≥1500ms Measured: ≥5.8A @ 1400ms


AUTO | Riverside (Summer) | OFF-PEAK |
8:25 AM | $0.220/kWh | SRC=GRID |
Load=752W | Fan=65%

Summary

Through the development of this system, our team encountered various forms of both software and hardware complexities. Development highlighted flaws in our design that were later rectified.

- Initial development begins on isolated modules to test basic functionality
- Manufacture of PCB to populate with required components

This PCB did not meet the standards needed for further development

- Replaced with perf-board with components populated instead

Failure of our initial design required replacement of all sensors

- Sensor modules placed on breadboard and separate from main PCB due to time constraints

Finalization of our software and modification for changes in sensor calibrations along with layout diffs.

This project provided emphasis on the importance of iterative design, along with our ability to adapt to changes in specification when found infeasible or otherwise unavailable.

Our team tested for fully autonomous operation with a priority on safe, and efficient switching. Ensuring that both the final product and demonstration meet the goals we set forth in our proposal.

Acknowledgements

Mohammadamir Kavousi: Suggestion for original design concept and primary inspiration for our project

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Video demo link: https://youtu.be/YzSE4Eos70A